

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2019 - 2020 (Even Semester)

Innovative Practices Description

Unit I Drive Characteristics

Degree, Semester & Branch	: VI Semester B.E. EEE
Course Code & Title	: EE8601 Solid State Drives
Name of the Faculty member	: Mrs. S. Jeyanthi
Name of the Topic	: Typical load torque characteristics
Name of the Innovative Practice	: Mini Map
Date & Duration	: 04.01.2020 & 15 Minutes

Description:

A mini map is a diagram used to visually organize information. A mini map is hierarchical and shows relationships among pieces of the whole. Mini mapping in a learning environment which requires an enhanced way of thinking, learning and presenting. Mini Mapping enhances the learning capacity in terms of number of ideas generated as well as improves presentation focus. Mini maps are used to generate, visualize, structure and classify ideas. Mini maps go under variety of names. They are also known as concept maps, semantic mapping, knowledge mapping, think links, graphic organizers or cognitive maps.

Learning Outcome:

- The students will be able to visualize the concepts and organize information.

Justification for choosing the topic using Mini Map activity:

Mini mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mini mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches.

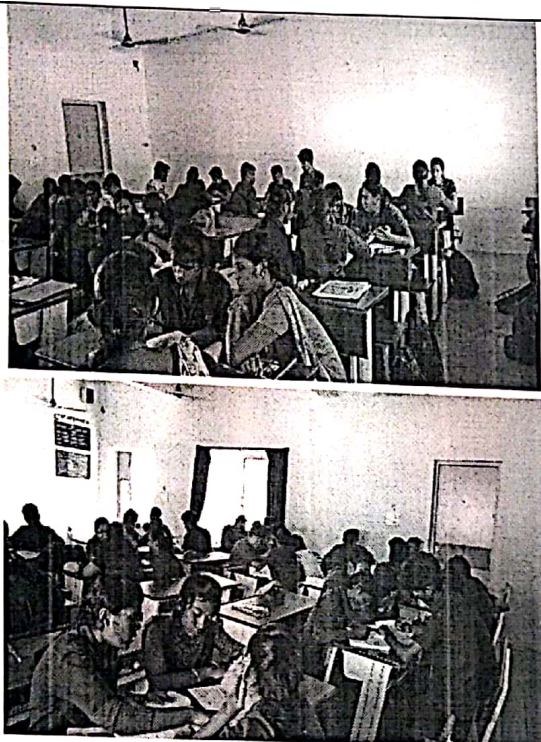
Implementation (Plan & Execution) with Proof:

Mini mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mini map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

In my subject Mini map was conducted for Unit-I Drive characteristics for the topic typical load torque characteristics. The students are voluntarily drawn the map. The sample copy of the mini map was done by the students was shown below.

Innovative Teaching Method Execution

Typical load torque characteristics – Mini Map



Load torque characteristics

Classification of load torque

- **Active load torque:** Load torque which has the potential to drive the motor under and equilibrium condition. Eg: torque due to tension.
- **Passive load torque:** Load torque which always "oppose the motion". Eg: torque due to friction.

Components of load torque

- **Friction torque (T_f)** present at motor shaft.
 - Vacuum: $T_f = B\omega_m$
 - Edmund: $T_f = C\omega_m$
- **Windage torque:** Wind generates a torque opposing the motion. $T_w = C\omega_m^2$
- **Mechanical work:** Nature of the torque depends upon the particular application.

$T_L = T_f + T_w + T_L$
 $= B\omega_m + C\omega_m^2 + T_L$

Nature of load torque

- **Constant torque:**
 - Eg: cutting
 - $T_L = K$
- **Linear load:**
 - Eg: Generators type
 - $T_L = K\omega_m$
- **Quadratic load:**
 - Eg: fan, rotary pump
 - $T_L = K\omega_m^2$
- **Other load:**
 - Eg: Lifting, boring etc

By: _____
 Group No: 10
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Reflective Critique:

Benefits:

The benefits of Mini Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mini map made slow learners to remember the information more quickly.

Challenges:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to draw the map to indicate relationships between the topics in typical load torque characteristics. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes to complete the whole topics in Unit-I.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	1	1	-	-	1	2	1	-	1

Reference:

1. Gopal K.Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, 1992.

CO1: After completion of the course, it is expected that students will be able to explain the steady state operation and transient dynamics of a motor load system.

S. Jeyaraj
6/1/2020
Signature of the faculty member

S. K.
06/01/2020
HoD/EEE

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2019 - 2020 (Even Semester)
Innovative Practices Description

Unit II Converter / Chopper Fed DC Motor Drive

Degree, Semester & Branch	: VI Semester B.E. EEE
Course Code & Title	: EE8601 Solid State Drives
Name of the Faculty member	: Mrs. S. Jeyanthi
Name of the Topic	: Applications of DC motor drives
Name of the Innovative Practice	: One Minute Paper
Date & Duration	: 27.01.2020 & 05 Minutes

Description:

Minute paper is a short writing activity within the classroom in which students generate answers in response to questions asked by the teacher regarding the concepts taught in class. These questions stimulate a student to reflect on the lesson taught and learned and provides feedback regarding their understanding which helps the teachers to plan ahead for the next class. Its major advantage is that it provides rapid feedback on whether the instructor's main idea and what the students perceived as the main idea are the same.

Learning Outcome:

- The students will be able explain the applications of DC motor drive in various industries.

Justification for choosing the topic using one minute paper activity:

Applications of dc motor drive in wide range of industries due to their constant speed operation, dynamic response and high starting current. Since, one minute paper activity provides a conceptual bridge between successive class periods. Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

Implementation (Plan & Execution) with Proof:

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

Innovative Teaching Method Execution

Applications of DC motor drive – One minute paper

Application of DC motor Drive:-

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are more applications of DC motor.
In older days mostly DC drive is used, some applications are

- * steel rolling mill
- * crane and hoist system
- * winding/coiler
- * Marine - DC generator power a feed
- * Mining

and above above applications are used to maintain the torque (or) for wide range of speed.

Applications:-

- * Steel Roll Milling
- * hoist & crane system
- * Mining
- * Extruding
- * cranes or
- * Marine System.

Applications of DC Motor drives:-

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(i) Rolling Mills:-

In rolling mills dc machines are used because of wide range of speed and torque.

(ii) Hoist or crane

(iii) Coil winding machine

(iv) Marine applications

(v) Mining applications - In mining they are used for wide range of speed at given torque

Reflective Critique:

1. Pre-implementation Reflection :

• Benefits:

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

• Challenges:

- In the class mostly boys hesitate to answer the questions.
- Time utilization for conducting activity.

2. Post-implementation Reflection :

• Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

• **Challenges:**

- Slow learners were not able to understand some topics during discussion hours.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	2	1	1	1	-	-	1	2	1	-	1

References:

1. Vedam Subramanyam, " Electric Drives Concepts and Applications ", 2e, McGraw Hill, 2016
2. N.K. De., P.K. SEN" Electric drives" PHI, 2012.

CO2: After completion of the course, it is expected that students will be able to examine the operation of converter/chopper fed dc drive.

S. Jeyaraj
21/11/2020
Signature of the faculty member

S. K 31/01/2020
HoD/EEE

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2019 - 2020 (Even Semester)

Innovative Practices Description

Unit III Induction Motor Drive

Degree, Semester & Branch	: VI Semester B.E. EEE
Course Code & Title	: EE8601 Solid State Drives
Name of the Faculty member	: Mrs. S. Jeyanthi
Name of the Topic	: Closed loop control of CSI drives
Name of the Innovative Practice	: Think Pair Share
Date & Duration	: 12.02.2020 & 15 Minutes

Description:

- T (Think): Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question. [Takes 2 Minutes].
- P (Pair): Teacher asks another question, related to the previous one, that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 2 Minutes].
- S (Share): Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 10 minutes].

Learning Outcome:

The students were able to explain conceptual understanding and analyse the pros and cons of voltage source inverter and current source inverter.

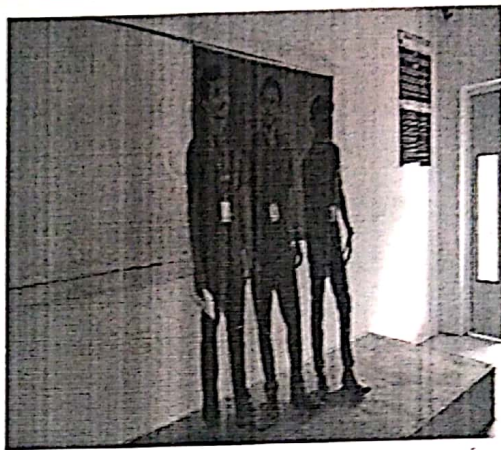
Justification for choosing the topic using Think-Pair-Share activity:

- It helps students to think individually about a topic or answer to a question.
- It teaches students to share ideas with classmates and builds oral communication skills.
- It helps focus attention and engage students in comprehending the reading material.

Implementation (Plan & Execution) with Proof:

Think-Pair-Share innovative practice conducted for III year EEE students, after explained the concept of voltage source inverter and current source inverter fed induction motor. First, I asked the students to think about the difference between VSI and CSI fed

induction motor for 2 minutes. Then I make them as a pair to discuss their neighbour's and asked the students to discuss about pros and cons of VSI and CSI fed induction motor for 3 minutes. Finally I asked the one of the team to explain the concept to whole class for further discussion. The students from group share their points and participated in the discussion for 10 minutes.

Innovative Teaching Method Execution	
VSI and CSI fed induction motor – Think Pair Share	
	
Advantages of VSI: • It's size is small • Output voltage constant • Low cost Disadvantages of VSI: • In VSI speed range full torque Advantages of CSI: • Full of current & power • More reliable • Output current constant • Short range full & power Disadvantages of CSI: • Slow speed • Costly • Efficiency < 50% <i>Team: Electronics H.M. Shrinagesh A.P. M. Manishkumar A.P. C. Mohanesh</i>	CSI - Output current is constant - Commutation circuitry required - Cost is high - Inductor is required for torque & current source - It will be operated in first two quadrant. VSI - Output voltage is constant - Commutation is not required - Cost is low - Inductor does not required. - It will be operate only one quadrant.

Reflective Critique:

Benefits:

Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating.

Challenges:

Four to six students are not shown interest to actively participate in groups. I planned the activity for 10 minutes. But in Class room it takes 15 minutes.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	2	2	1	1	-	-	1	-	1	-	1

References:

1. Gopal K.Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, 1992.

CO3: After completion of the course, it is expected that students will be able to apply the control methods for the induction motor drives for suitable applications.

S. Sarda
18/2/2020
Signature of the faculty member

S. K. K.
18/02/2020
HoD/EEE

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2019 - 2020 (Even Semester)

Innovative Practices Description

Unit IV Synchronous Motor Drive

Degree, Semester & Branch : VI Semester B.E. EEE
Course Code & Title : EE8601 Solid State Drives
Name of the Faculty member : Mrs. S. Jeyanthi
Name of the Topic : Three phase current source fed synchronous motor
Name of the Innovative Practice : Visual Clicker
Date & Duration : 04.03.2020 & 25 Minutes

Description:

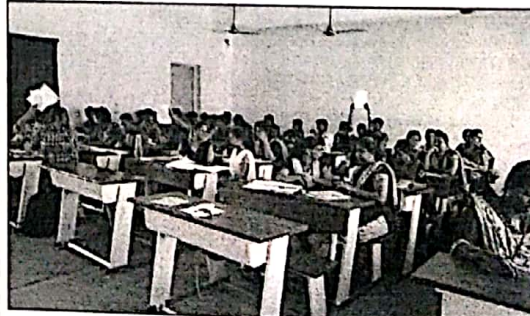
A Quiz is usually a form of a student assessment, but often has fewer questions of less difficulty and requires less time for completion than a test. A quiz is a form of game to test the students' knowledge about a certain subject. A brief assessment used in education and similar fields to measure growth in knowledge, abilities and skills.

Learning Outcome:

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam.

Implementation (Plan & Execution) with Proof:

- ✓ First of all the students are divided into 8 small teams
- ✓ Option (A,B,C,D) Cutouts are given to each group
- ✓ Then I will display the Multiple choice question (MCQ) by using LCD projector at duration of 2 minutes.
- ✓ From each team any one of the student show the corresponding answer cutout.
- ✓ After that correct answers are discussed in class room itself.

Innovative Teaching Method Execution	
Three phase current source fed synchronous motor – Visual Clicker	
	

Reflective Critique:

Benefits:

- ✓ Due to this activity the students are actively participated in group learning skill.
- ✓ Each student to improve their remembering capability.
- ✓ Easy to revise the fundamental concepts.

Challenges:

- ✓ Few students are not interested to hold the answer cut outs.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	2	1	1	1	-	-	1	-	1	-	1

References:

1. Gopal K. Dubey, Fundamentals of Electrical Drives, Narosa Publishing House, 1992.
2. J. Granavadivel, Karthikeyan & Santhi, A Textbook of Solid State Drives, Anuradha Publications, 2015 Revised Edition.

CO4: After completion of the course, it is expected that students will be able to analyze the operation and performance of synchronous motor drives.

S. K. 9/5/2020

Signature of the faculty member

S. K. 9/5/2020
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